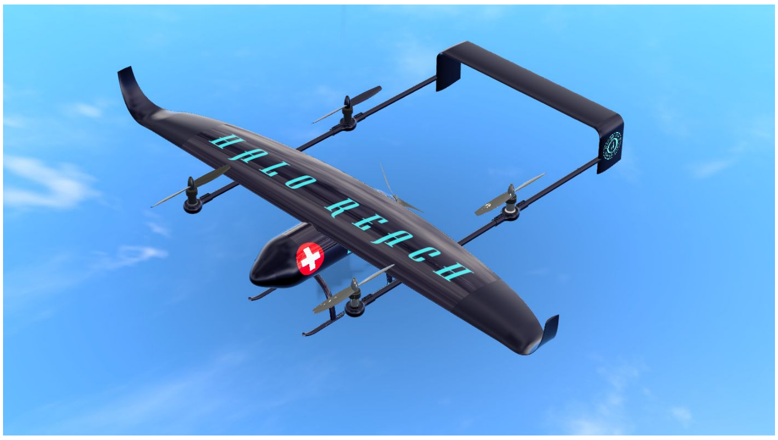
# HALO REACH: Navigation, Communication and Data Relay System

UAS for Disaster Relief - AE3 2024

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**Abstract**

This report aims to detail the development process and performance evaluation behind selection of the Navigation sensors, Guidance subsystem algorithms, Communication and Data Relay System of HALO REACH. To this end, an in-depth literature review and trade-off studies were used to select the suitable Navigation Sensors and Guidance framework suitable for REACH. To address the Communication and Data Relay System, a novel approach was taken to model signal losses and select suitable hardware.

The selected Navigation sensor components were the OEM7700(GNSS module), Novatel PwrPak7(GNSS antenna), SBG Ellipse 2(IMU), DPS310(Altitude Barometer), Ouster OS1(Lidar) and FliR Duo Pro R 640(Camera). The selected Guidance System workflow was YOLO CNN for object detection, Octomap for Terrain Representation, RRT* for Trajectory Planning and Dubin's Path for Trajectory Smoothing. Both the Navigation sensor selection and Guidance algorithm framework suggested were selected for their reliability and accuracy. The MILLI SAT terminal and an Intelsat Fleet were selected to enable BLOS communication over a 185km radius. The system was designed successfully with a sufficient link margin of 8dB and bitrate of 1.26Mbps. The Data Relay designed for challenging high-rise urban environments and employed the Teltonika RUTX09 router and MIMO WMM86-7-27 directional antenna to successfully provide coverage with sufficient signal strength to a 19.6$m^2$ area. However, data rate requirements were not met with capabilities to only provide sufficient bandwidth for 22 users as opposed to 50.

# Contents

## List of Figures

# List of Tables

# Nomenclature

| | |
|---|---|
| $\beta_{max}$ | Max Spectral Efficiency(bps/Hz) |
| $\gamma_r$ | Specific Attenuation(dB/km) |
| $\mu$ | Mean(dB) |
| $\phi$ | Longitude(°E) |
| $\pi$ | Pi(3.14159) |
| $\psi_{elev}$ | Elevation Angle(°) |
| $\sigma$ | Standard Deviation(dB) |
| $\theta$ | Latitude(°N) |
| $\zeta$ | Localisation variability for Log Normal model(dB) |
| $AdLoss$ | Additional Loss(dB) |
| $AtmLoss$ | Atmospheric Loss(dB) |
| $b_r$ | Estimate for number of buildings blocking LOS view |
| $B_W$ | Bandwidth(Hz) |
| $c$ | Speed of Light ($3\times10^8 m/s$) |
| $d$ | Slant Range(m) |
| $EIRP$ | Effective Isotropic Radiated Power(dBW) |
| $f$ | Frequency(Hz) |
| $FSPL$ | Free Space Path Loss(dB) |
| $G_r/T$ | Receiver Gain to Noise Temperature Ratio(dBi/K) |
| $G_t$ | Transmitter Gain(dBi) |
| $h_0$ | 0°C Mean Isotherm Height (km) |
| $h_{GEO}$ | GEO Satellite Altitude(35786km) |
| $h_{R_x}$ | Receiver height(0km) |
| $h_r$ | Isotherm height(km) |
| $h_{T_x}$ | Transmitter height(0.5km) |
| $K_B$ | Boltzmann Constant(-228.6 dBW/K/Hz) |
| $L_{LOS}$ | LOS Loss(dB) |
| $L_{NLOS}$ | NLOS Loss(dB) |
| $L_s$ | Shadowing Loss(dB) |
| $L_{TOTAL}$ | Total Loss(dB) |
| $P(LOS)$ | Probability of Line of Sight Connection |

| | |
|---|---|
| $P_{extra}$ | Extra Power Required(W) |
| $P_t$ | Transmitter Power(dBm) |
| $R_{0.01}$ | Rainfall Coefficient(mm/hr) |
| $R_e$ | Radius of the Earth(6371km) |
| $RSSI$ | Received Signal Strength Indicator(dBm) |
| $SNR$ | Signal to Noise Ratio(dB) |
| AoA(°) | Angle of Attack |
| BLOS | Beyond Line of Sight |
| BR | Bias Repeatability |
| BS | Bias Stability |
| FOV(°) | Field of View |
| GCS | Ground Control Station |
| GEO | Geostationary Earth Orbit |
| GNSS | Global Navigation Satellite System |
| GTOW | Gross Takeoff Weight (kg) |
| HALO | Humanitarian Aid Logistics Operations |
| IMU | Inertial Measurement Unit |
| LEO | Low Earth Orbit |
| LOS | Line of Sight |
| REACH | Rapid Emergency Aid and Communications Hub |
| ROM(°) | Range of Motion |
| RW | Random Walk |
| SATCOM | Satellite Communication |
| UAS | Uncrewed Aerial System |
| VTOL | Vertical Take-Off and Landing |

# 1 Introduction

Deployment of UAVs can be vital in mitigating the aftermath of disasters[1]. HALO's response to the design brief [2] is REACH, a fully autonomous drone aiming to revolutionise disaster relief efforts, by boasting versatile payload and data relay functionalities. From a Systems perspective, design of a robust Guidance and Navigation system was paramount in ensuring safe completion of missions. Furthermore, the 185km mission radius was unconventionally large for a drone with REACH's lower GTOW and mission profiles, which posed significant challenges in the design of lightweight Communication and Data Relay Systems.

This report aims to select suitable Navigation sensors and guidance algorithms that ensure safe completion of REACH's mission profiles autonomously, prioritising reliability at every stage. In addition, the large mission radius called for a novel approach to Communication system design to complete missions worldwide based on our customer's needs [3]. Furthermore, a novel approach was taken for lightweight data relay design to operate even in the most challenging high-rise urban environments desired by our customers. Systems design was a collaborative effort, reports from Lin [4] and Aggarwal [5] should be read together for a holistic understanding of the GNC subsystem design in figure 3.

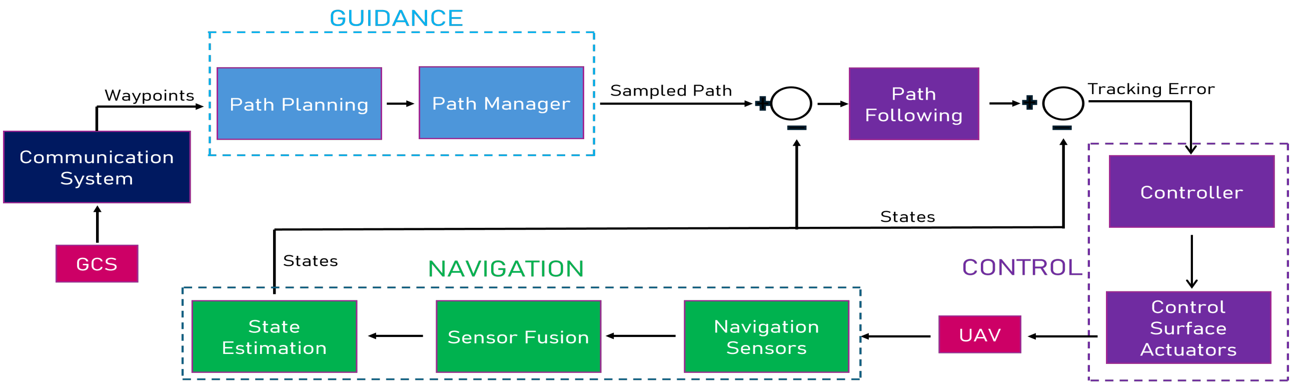


**Figure 3.** GNC Subsystem adapted from [6]

# 2 Navigation Sensor Selection

Navigation sensors are crucial in measuring the navigational states required for safe completion both missions profiles and for designing controllers developed by the FDC sub-team. The approach taken for this section was to first introduce the type of sensor selected. Then an example of the selection process is explored in detail for the GNSS module, arriving at a final hardware choice. To avoid repetition the final choices for the other sensors are then presented with extra detail being added where necessary. Finally, measures to ensure redundancy is explored to further enforce reliability.

Given the use of VTOL and autonomous optical navigation(required by brief[2]), accurate linear and angular position/velocity state measurement was essential. The combined states required by both the FDC sub-team and navigation system were the 12 flight dynamics states[7]. Appendix A lists the sensors typically used for UAVs and their respective measured states.

The final selection of sensors included GNSS, IMU, an Altitude Barometer, Lidar and an Optical/Infra-red camera. Both the GNSS and IMU are fundamental in precise geographical navigation and maintaining stability and are always present in UAVs. As a result of VTOL controller requirements, the altitude measurement needed increased accuracy from that a standalone GNSS would provide, hence an altitude barometer was used[8]. Lidar was used to enhance distance accuracy of obstacle in obstacle detection algorithms, especially in low visibility conditions. Optical navigation was facilitated by a camera with both optical and infra-red capabilities to offer versatility in nighttime missions. The AoA wind vane is only used on high performance orientated drones (eg.military) and thus was excluded for out case. Although airspeed sensors are commonly used, [9] suggests with advanced sensor fusion architecture (investigated by Aggarwal [5]) an airspeed sensor isn't needed. Furthermore, airspeed sensor blockage is a common cause of drone control system malfunction and crash [10], so reliance on one was avoided. All other sensor options are excluded as their measured states can be obtained to a sufficient level of accuracy from already selected sensors that provide better weight and size characteristics.

## 2.1 GNSS Module Selection

In order of importance, the main selection criteria for the GNSS module was reliability (eg.signal coverage, environmental resistance), accuracy and size/weight. Note that as a result of the ability to recharge the battery, considerations of power usage were not as important.

To ensure reliability, vibration and weather resistance was necessary to maintain data measurement integrity [10]. Only sensors that adhered to MIL-STD-810[11] and IP67[12] or above were considered for vibration and weather resistance respectively. Both these standards, were selected as suitable for the mission profile and purpose of REACH. Utilising multi-frequency GNSS was the best way to ensure high reliability. It ensured that multiple satellite constellations could be used, improving coverage and signal strength. Positional accuracy and convergence time was shown to improve when switching to multi-frequency from single frequency, reducing by a factor of 50 and 2 respectively [13]. The cumulative effect is that REACH could be deployed faster and perform geolocation navigation more accurately. To this end, only GNSS modules that utilised the L1/L2/L5 frequency band (able to use all constellations) were considered. For accurate GNSS based navigation, 20Hz was the lowest acceptable update rate[10] and position accuracy above $\pm 1.5$m was desired.

Figure 12 in Appendix A shows options considered for the GNSS module. The best and final choice was the OEM7700[14], based on our previously mentioned requirements.

## 2.2 Final Sensor Selection

Following a similar process and considering the same selection criteria as section 2.1, figures 13-16 in Appendix A details the options considered for each component. The final hardware selections are shown below in table 2. IMU considerations included selecting a Quartz MEMS IMU that significantly reduced BS,BR and RW which together account for the error in long and short term measurement stability [15]. The selected ROM of the camera and mount combination, enhanced aspects of the optical navigation system discussed in section 3, by improving moving object tracking capabilities. It also allowed image capture directly below the UAV, required when assessing safety of landing in the payload mission. The importance of high resolutions in altitude barometers is explored in detail in [8].

**Table 2.** Final Sensor/Component Selection [16] [17] [18] [19] [20] [21]

| Component | Name | Mass (g) | Dimensions (mm) | Power (W) | Other Factors |
|---|---|---|---|---|---|
| GNSS Antenna | Novatel PwrPak7 | 500 | 147x125x55 | 3.2 | N/A |
| IMU | SBG Ellipse 2 | 10 | 27x19x10 | 0.4 | BS, BR, RW |
| Alt Barometer | DPS310 | 1 | 2x3x1 | 0.1 | Resolution ($\pm$0.02m) |
| Lidar | Ouster OS1 | 447 | 7x3x3 | 1.2 | FOV (H:360°,V:45°) |
| Camera | FLIR Duo Pro R 640 | 325 | 87x85x69 | 10.0 | Resolution (4000x3000) |
| Camera Mount | 3DR Tarot T-2D | 200 | Unavailable | N/A | ROM (360° pan, 90° tilt) |

No hardware redundancies were used as the software redundancies explored by Aggarwal [5] offered greater flexibility, cost-effectiveness and lower weight. Modern advancements in software based fault management enhance overall reliability without the need to duplicate physical components, thereby also reducing maintenance requirements[22].

# 3 Guidance System

Note that the following section is literature heavy and this report covers the selection process of algorithms/software rather than a holistic explanation of each one. Therefore the reader is encouraged to explore the provided references where deemed necessary.

Obstacle detection and avoidance algorithms were vital as part of the Guidance system in figure 3 to ensure REACH could identify and react to obstacles unaccounted for in data sent from the GCS. Figure 18 in Appendix B highlights this in a detailed example of how this algorithm would be used in the payload mission profile. Obstacle detection and avoidance algorithms are divided into 4 stages:

1. Obstacle Detection (Converting signals from sensors into mapable data)
2. Terrain representation (Mapping the 3D space, including obstacles and UAV itself)
3. Trajectory Generation (Generating collision-free sequence of waypoints around obstacles)
4. Trajectory Smoothing (Joining waypoints with exact trajectory to follow)

## 3.1 Object Detection

Hardware used for object detection is divided into the categories outlined in table 17 in Appendix **??**. The fusion of the stereo vision FLIP Duo camera and non-visual Lidar combines the advantages of both, being robust to varying light conditions and electromagnetic interference[23]. Traditional detection algorithms include edge detection (Canny[24]), corner detection (Shi-Tomasi[25]) and feature matching (SIFT[26]). Modern detection approaches include CNNs like YOLO[27](<45fps), SSD[28](<21fps) and Faster R-CNN[29](<7fps). Use of a CNN was selected due to superior accuracy, robustness to varying light conditions and adaptability to diverse object appearances[30]. Amongst CNNs, YOLO is optimised for real time performance having the best fps and therefore was selected for REACH. Furthermore, YOLO has been established suitable for use in UAVs[31].

## 3.2 Terrain Representation

To allow REACH to maintain situational awareness of its surroundings, selection of a framework was necessary that allowed for efficient storage and querying of the 3D space, via use of a hierarchical structure. As REACH performs its mission the 3D terrain will be updated with new sensor data and queried to identify collision risks. Commonly used frameworks for this application include Octomap[32], Voxblox[33], RTAB-Map[34]. Octomap was selected due to its probabilistic nature making it the option most robust to sensor noise, especially common in REACH's mission profile. This is because each voxel stores a probability of an object existing as opposed to a boolean value. Moreover, Octomap offered the best balance between computational efficiency and accuracy, essential when implementing real-time detection algorithms.

## 3.3 Trajectory Planning

The framework to detect and map obstacles to a 3D space has been explored. Now, the options of algorithms to generate collision free trajectories are considered. Amongst the options suitable for dynamic environments there were 3 main choices: A* algorithm, RRT* algorithm and reinforcement learning[35]. Reinforcement learning is a relatively unexplored realm that requires extensive computational resources to train resources without a significant increase in accuracy and hence was excluded[36]. The A* algorithm was excluded due to efficiency of generated path being dependent on a heuristic function which in most cases is hard to determine. The RRT* algorithm was the final choice as a result of its balance of computational efficiency and accuracy that made it the best choice for REACH's high dimensional environment[37]. RRT*'s incremental construction provided unparalleled adaptability to reacting to new obstacles. Furthermore, RRT* is an asymptotically optimised algorithm where solutions improve with more iterations[38].

### 3.4 Trajectory Smoothing

Suitable methods of generating trajectories from a sequence of waypoints can be restricted to choice between Dubin's path[39](shortest distance with minimum feasible turning radius considerations) or some form of polynomial interpolation between waypoints[40][41]. As REACH is a fixed wing UAV, it exhibited maneuvering constraints and so Dubin's path was a natural choice to account for this problem.

### 3.5 Algorithm Testing and Verificiation

Before deployment of the complete algorithm to REACH, simulation[42] followed by field testing[43] will occur. To address the former, Unreal engine's virtual environment is an industry standard tool often used to test autonomous systems before deployment[44]. Its high fidelity simulation environment would be used to mimic the obstacles and terrain conditions REACH would be likely to encounter in its mission profiles. The algorithm would be implemented within the virtual environment and performance metrics such as detection accuracy and path optimality will be analysed before moving onto field testing. Field tests will involve assessing performance of the algorithm in incrementally complex scenarios (eg.adding more complex obstacles). Then the sensor data and video feed will be analysed to again quantify detection accuracy and path optimality until performance is deemed acceptable. The bottleneck of the selected algorithm is the use of YOLO, which is why quantifying detection accuracy at each stage is essential for successful deployment.

## 4 Communications System

As per the requirements detailed in the Introduction, a system was required to relay information with a Ground Control System (GCS) such as telemetry data, low quality video feed and pilot commands. However, due to the constraints of a 185km mission radius [2] from the design brief choices were limited to design of Beyond-Line-Of-Sight (BLOS) communications systems, eliminating the possibility of use of Radio-wave systems commonly used for this application. Thus this section details the design of a Satellite Communication System (SATCOM), the most widely adopted BLOS system for UAVs.

SATCOM systems consist of a CGS satellite terminal communicating with the satellite terminal on the UAV, using a satellite as an intermediate. Data links from a satellite terminal to the satellite are referred to as the uplink and the opposite is referred to as the downlink.

Design of the CGS and its hardware is detailed by Lin[4]. In contrast to the GCS satellite terminal, the UAV satellite terminal was constrained in terms of size and weight, thus numerical analysis was done to obtain suitable hardware that met our requirements and constraints. The following steps outline the design process used for selecting the UAV SATCOM system:

1. Evaluate performance requirements of UAV-Satellite data link (eg.data rates)
2. Select suitable frequency band
3. Select suitable Satellite fleet
4. Obtain required parameters for robust communication
5. Select suitable transmission/receiving hardware

The analysis used to perform Step 4 is known as a Link Budget Calculation, a widely used tool in Telecommunications design [45]. Throughout this section a conservative design procedure was adopted, allowing for sufficient safety margins that ensured a final design, robust to unpredictable conditions.

### 4.1 Performance Requirements

It is unlikely that HALO Reach will encounter military threats such as signal jamming and interception in humanitarian disaster relief efforts, therefore such design considerations were ignored. This reduced design of the SATCOM system to consideration of the following main factors:

- **Data Rates**: Data links must support the data rate requirements for various types of transmitted data. Table 3 presents conservative upper range estimates for the data rates and bandwidth of typical data transmitted across the link.
- **Signal Strength**: Ensuring reliable transmission and reception of signals is crucial, even over long distances. This reliability is essential because some data, such as pilot commands and critical telemetry, is vital and failure to receive this data could result in the loss of the UAV.
- **Adverse Weather Conditions**: The communication system must still be able to be received in the event of unpredictable signal deteriorating weather conditions.

**Table 3.** Data Rate and Bandwidth Requirements based on [46]

| Function | Bitrate (Mbps) | Bandwidth (MHz) |
|---|---|---|
| Control Commands | 0.05 | 0.10 |
| Telemetry data | 0.10 | 0.20 |
| Low quality video feed | 1.00 | 1.00 |

All the main factors mentioned are accounted for in Section 4.4. The data in Table 3 was further divided into downlink(control commands) and uplink(telemetry data and low quality video feed) data. Clearly, the uplink data is far more bitrate/bandwidth intensive and design of this more constraining uplink ensured both the uplink and downlink requirements were met. This imposed minimum performance limits of 1.10Mbps bitrate and 1.20MHz bandwidth.

### 4.2 Frequency Band Selection

The most commonly used frequencies for UAV SATCOM Systems range from 12-40GHz, comprising of the Ku(12-18GHz), K(18-26GHZ) and Ka(26-40GHz) bands [47]. Operation within the Ku band was selected as it has suitable bandwidth availability for our 1.20MHz requirement. Selection of the lowest frequency band that met the performance requirements had the additional benefit of having the best atmospheric penetration power, which resulted in lower atmospheric and propagation losses. Propagation losses was found to be the most significant factor that deteriorates signal quality in Section 4.4 . Furthermore, this reduction in atmospheric loss allowed for use of smaller, lightweight antennas that were more suitable for integration on a UAV. The frequency selected was 14.5GHz and this was the value used for subsequent calculations. It is worth noting that the actual frequency allocation will vary around 14.5GHz in different regions which is governed by the International Telecommunications Union (ITU) [48]

### 4.3 Satellite Fleet Selection

Requirements for satellite fleet selection were the following:

- **Worldwide Ku band Coverage**: Reliable coverage at our chosen frequency band is important for signal strength performance requirements. This allows for disaster relief operations worldwide.
- **Geostationary (GEO)**: Continuous and stable coverage that doesn't change all year round eliminates the requirement for satellite tracking software to adjust antenna direction as the satellite moves across the sky.

- **UAV Applications**: Satellites that have been used successfully in UAV applications have demonstrated their reliability and performance under real-world conditions. This reduces the uncertainty and risk associated with deploying a communication system. Moreover, this ensures compatibility with existing UAV communication hardware reducing integration time and efforts.

A satellite fleet consisting of 6 evenly spaced IntelSat GEO Satellites [49] that provided worldwide coverage was selected for this application and are included in Appendix C. Their locations on the world map are indicated in figure 4. The Intelsat provider also offers training programs and case studies to aid UAV engineers in utilising the satellites effectively, even offering dedicated support teams that can provide specialised assistance when required.

## 4.4 Link Budget Analysis

Link Budget calculations are a means of calculating the safety margin of receiving a signal, known as the link margin and also serve the purpose of verifying the bit-rate requirements in dection 4.1 are met for our data link. From this a suitable SATCOM terminal was selected powerful enough to meet an acceptable safety margin and support the bitrate requirements.

The design requirements called for flexibility of worldwide operation, however to reduce complexity, the operating domain was restricted to between 75°N and −60°N latitude (pink lines in figure 4), ignoring widely uninhabited areas beyond these bounds. The subsequent calculations were computed using MATLAB for every point in the operating domain.

### 4.4.1 Free Space Path Loss

The FSPL for each point was calculated using the Friis transmission method in equation (1) from [45] and accounts for the loss of signal strength as it propagates through free space. The slant range is the straight line distance between UAV and satellite and was computed with equation (2), using the longitude and latitude of the domain points as well as the longitudinal data of the closest satellite that provided coverage from table 8 in Appendix C. The elevation angles (angle between horizontal plane at UAV location and line of site to satellite) was computed using equation (3) and lie in between 0° and 90° for all domain points.

$$\text{FSPL} = 20\log_{10}\left(\frac{4\pi d f}{c}\right) \tag{1}$$

$$B = \phi_{\text{UAV}} - \phi_{\text{SAT}} \qquad b = \arccos(\cos(B)\cos(\theta_{\text{UAV}})) \qquad d = \sqrt{R_e^2 + h_{\text{GEO}}^2 - 2R_e h_{\text{GEO}}\cos(b)} \tag{2}$$

$$\psi_{elev} = \arccos\left(\frac{h_{\text{GEO}}\sin(b)}{d}\right) \tag{3}$$

### 4.4.2 Atmospheric Loss

Atmospheric losses account for the loss in signal strength due to absorption by different atmospheric molecules, mainly oxygen and water. The atmospheric loss for frequencies greater than 10GHz is almost always dominated by attenuation due to traversing clouds and rain (Rain fade) [50]. Thus Rain Fade losses was the only factor accounted for. The globally adopted ITU Rain Fade model was chosen due to its robust data support and compliance with regulatory standards. In contrast, models such as the Crane Model, which are less frequently utilized, tend to be less accurate and reliable.

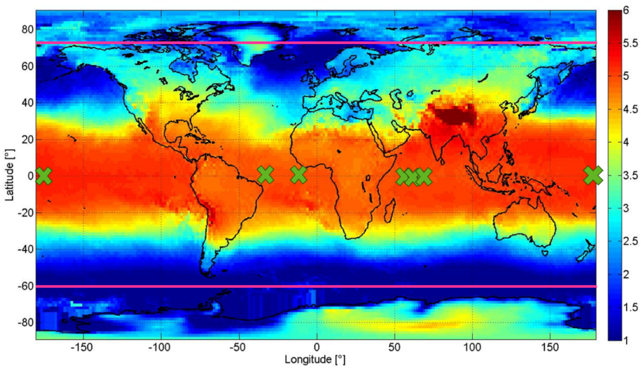


**Figure 4.** Worldwide 0°C Isotherm Height Map

The specific attenuation was calculated using equation (4) where $k$ and $\alpha$ are frequency dependent parameters using [51]. At each domain point

$R_{0.01}$ was obtained from [52]. Utilising $R_{0.01}$ ensures that there was a 99.9% confidence level that Rain Fade Loss wouldn't exceed the calculated value, thus providing a significant safety margin for reliable communication under various rain conditions.

$$\gamma_r = k{R_{0.01}}^{\alpha} \quad (4) \qquad h_r = h_0 + 0.36 \text{ km} \quad (5) \qquad \text{AtmLoss} = \gamma_r L_E \quad (6)$$

The $h_0$ for each domain point was obtained using figure 4 and $h_r$ was calculated using equation (5) from [53]. The calculation to obtain $L_E$, a correction factor depending on factors like $d$ and $\phi$ was highly involved and omitted from this report. Further details on this step can be found in [50]. Finally, atmospheric loss was calculated using equation (6).

### 4.4.3 Link Budget Calculation

The SNR is a way of quantifying the strength of a received signal and was calculated for every point using equation (7). The value of $G_r/T$ depends on satellite fleet hardware and is a constant. FSPL and AtmLoss were accounted for in sections 4.4.1 and 4.4.2. AdLoss was assumed to be negligible as per the ITU model and $B_W$ is the minimum value in section4.1. The EIRP term measures the strength of the transmitting hardware and was identified as the only factor dependent on the UAV SATCOM terminal selection. Therefore, by varying the EIRP within the typical UAV hardware range of 20-60 dBW in the subsequent calculations, the exact value that met the performance requirements was determined, allowing for the selection of suitable hardware.

$$SNR = \text{EIRP} + \frac{G_r}{T} - \text{FSPL} - \text{AtmLoss} - \text{AdLoss} - B_W - K_B \qquad \text{EIRP} = P_t + G_t \quad (7)$$

Link Margin quantifies the safety margin of the signal being received and was calculated for each point using equation (8). Bitrate was calculated using equation (9). $\text{SNR}_{\text{min}}$ and $\beta_{\text{max}}$ represent physical limits imposed by the QPSK 2/3 modulation scheme used by the Intelsat fleet [54]. $\text{SNR}_{\text{min}}$ sets a lower bound on signal strength required for reception and $\beta_{\text{max}}$ defines an upper limit on the bandwidth efficiency of the satellite, thereby constraining the achievable bitrate. Table 4 contains a summary of the parameters used to perform this analysis.

$$\text{LinkMargin} = \text{SNR} - \text{SNR}_{\text{min}} \quad (8) \qquad \text{Bitrate} = \min \begin{cases} B_W \log_2(1 + \text{SNR}) \\ B_W \beta_{\text{max}} \end{cases} \quad (9)$$

To ensure worldwide operation, the most challenging location with the lowest SNR was evaluated for sufficient LinkMargin and Bitrate, which ensured reliable operation in all other locations as well if this was the case. This location is detailed in table 5 and was found to be in China, a key target market identified by Moreno[3].

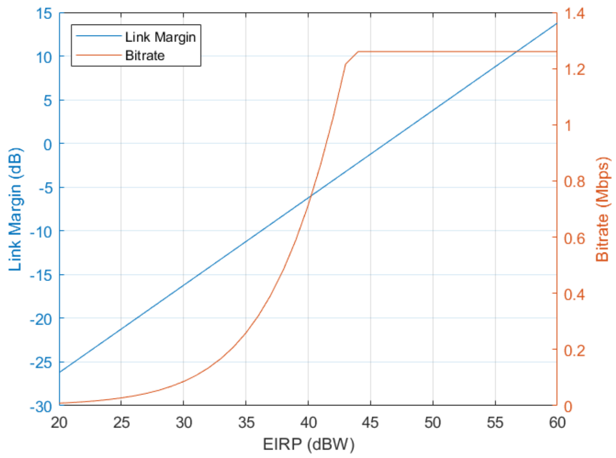


**Figure 5.** Link Margin and Bitrate vs EIRP

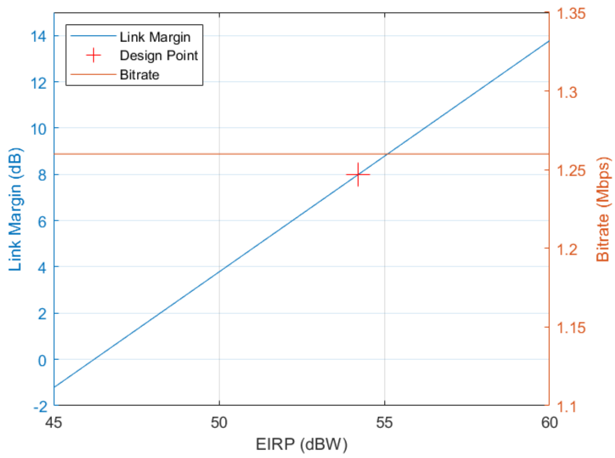


**Figure 6.** EIRP Design Point

Figure 4 shows that only above an EIRP of 46dBW will the signal be received and LinkMargin be greater than 0. Literature suggests a LinkMargin of roughly 4-8dB for robust communication [55], thus the conservative 8dB was selected as seen in figure 6, corresponding to roughly an EIRP of 54dBW. At this design point the bitrate is capped at the satellite hardware limit of 1.26Mbps, which satisfies our minimum 1.2Mbps constraint. Table 5 offers insight into the relative contribution of the different components of the link budget calculation so was included.

| Parameter | Value |
|---|---|
| $f$ | 14.5GHz |
| $B_W$ | 1.2MHz |
| $G_r/T$ | 7.0dB/K |
| $\text{SNR}_{\text{min}}$ | 3.3dB |
| $\beta_{\text{max}}$ | 1.05bps/Hz |

**Table 4.** Parameters for Link Budget Calculation

| Parameter | Value |
|---|---|
| Longitude | 105°E |
| Latitude | 40°N |
| FSPL | 197.50dB |
| AtmLoss | 20.23dB |
| SNR | 11.08dB |

**Table 5.** Parameters for Most Constraining Location at Design EIRP

#### 4.4.4 SATCOM Terminal Hardware Selection

The maximum EIRP of typical UAV SATCOM terminals was found to be 46dBW from market analysis. However, it is possible to increase the EIRP of a transmitter by increasing the power input, which scales exponentially with the deficit in EIRP. This is demonstrated in equation (10). The 10kg MILLI SAT H LW terminal was selected as it had the highest EIRP, whilst having significantly lower mass than other systems with similar capabilities. It's minimum power was 250W at a terminal EIRP of 45.7dBW, which following equation (10) yielded 256.8W of power. The system also has the capabilities of rotating azimuth 0 - 360° and elevation -8-90° so is capable of successfully tracking the position of the satellite as the UAV moves.

It is worth considering at this stage about the possibility of fitting different SATCOM terminals to the UAV at the GCS before operation based on mission location and weather conditions. Based on conversations with Modular Add-Ons and literature review [56], the 10 minute window from the design brief [2] for changing payload, prelight checks etc. was deemed too stringent to perform a communication system swap in addition. Furthermore, [57] explores in detail the increase in human error that can occur worker under rushed conditions compromising operational reliability. Therefore, the decision was made to opt for a 'one size fits all' solution and fit all UAVs with the MILLI SAT.

$$P_{\text{extra}} = 10^{\frac{\text{EIRP}_{\text{design}} - \text{EIRP}_{\text{terminal}}}{10}} \tag{10}$$

## 5 Data Relay System

The design process was largely similar to that of the Communications System in section 4. However, the previous analysis differs in factors considerations in the loss calculations as a result of the signal challenges when working in urban environments. We again opted for a conservative approach and designed for operation in the most challenging environment - a dense high rise urban setting (eg.Seoul).

### 5.1 Performance Requirements

Based on the Great East Japan Earthquake case study[3] outlined by Moreno [3], the following design requirements were obtained:

- **Coverage Area -** 19.6 **km**$^2$: This area was sufficiently large to be effective in the aforementioned case study and other large urban cities.
- **Frequency - 4G**: Mobile phones were amongst the worst impacted in signal outages, therefore a frequency accessible to both civilians and ground relief operations was required.

- **Data rate - 5000 kbps**: The system should be able to facilitate up to 50 people at any time sending emergency data (location,text etc.) estimated at 100kbps. Priority of the relay's bandwidth will be given to relief operations.

## 5.2 Path Trajectory

The altitude of the trajectory was fixed at 500m as the aerodynamic performance was optimised for operation at this height and deviation from this would negatively impact endurance times. This simplified the path trajectory problems to 2 dimensions. Typical 4G data relay ranges don't exceed 20km in range [58] which posed significant challenges for the 185 km mission radius requirement. However, the communication system designed in section 4 overcome this range limitation and facilitated data relay from ground users to the GCS using an ethernet cable between the 4G router and SATCOM terminal. This can be accommodated with an ethernet cable between the router and SATCOM terminal. Consequently, using the SATCOM terminal allowed for trajectories closer to the ground users, improving data rates without concerns over GCS communication distance.

There exists a significant amount of research into optimising UAV trajectory based on minimising a cost function of energy usage divided by received signal strength of ground users. The topic for most recent research like [59] is related to multi-UAV data relay systems. However, most single UAV data relay systems like [60], [61] rely on the specific location of ground users being known in order to optimise trajectories via non linear solvers and the previously mentioned cost function on MATLAB. However, these methods fell short for our specific scenario where the location and number of ground users are unknown and may vary within our coverage area.

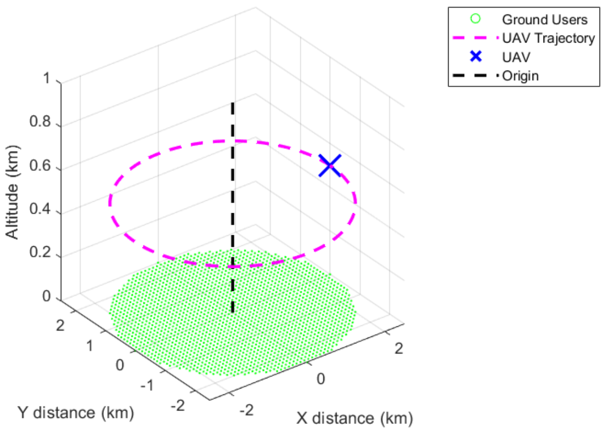


**Figure 7.** Trajectory 1: Radius-2.5km, Bank Angle-1.8°

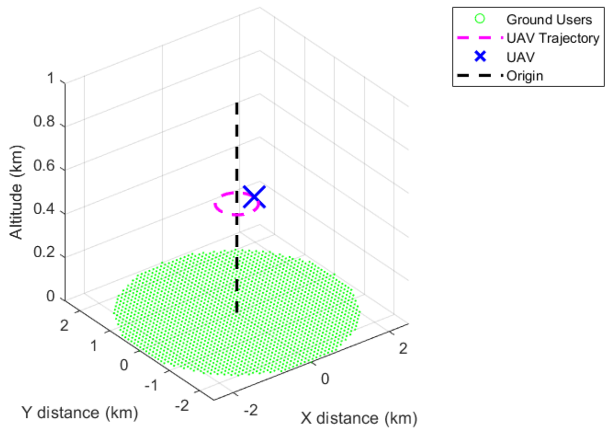


**Figure 8.** Trajectory 2: Radius-0.4km, Bank Angle-10°

As per section 5.1, coverage must be provided at all times to the whole area, therefore a circular trajectory is the most logical choice. The added benefit was that we can simplify the problem to consider only a single instance in time, as the whole geometry is axisymmetric. Mobile data relay antennas are either directional or omni-directional, with directional antennas being far more powerful. What was not immediately obvious was whether a trajectory of the radius of the coverage area with a directional antenna or a smaller radius trajectory with an omni-directional antenna would be better, with both hypothetically providing signal to the whole coverage area. Therefore, both trajectories were considered for the following analysis until the solution became clearer and can be seen in figures 7 and 8. The trajectory for the smaller path was determined using the maximum recommended sustained bank angle from [61].

## 5.3 Loss Calculation

Unlike the communications discussed in Section 4.4.3 , where a direct line-of-sight (LOS) was assumed, transmitting and receiving data from ground users involves overcoming obstacles such as buildings, which introduce additional shadowing losses. To this end, the Universal ITU stochastic model was implemented from [61] and equation (11) was used to estimate the probability of a LOS connection for all points in the domain. This estimation required empirical parameters $\alpha_b$, $\beta_b$, and $\gamma_b$ derived from building data that were set to 0.650, 900.88 and 30 respectively. These values, sourced from the city of Seoul [61], are more restrictive than typical high-rise urban data, making them more challenging to work with.

$$P(\text{LOS}) = \prod_{i=1}^{b_r} \left[ 1 - \exp\left( - \frac{\left( h_{T_x} - \frac{(i+0.5)(h_{T_x} - h_{R_x})}{b_r} \right)^2}{2\gamma_b^2} \right) \right] \quad \text{where} \quad b_r = \text{floor}\left( r_x \sqrt{\alpha_b \beta_b} \right) \tag{11}$$

Equation (12) is used to calculate the loss in the case of LOS and NLOS connections [62]. FSPL was calculated as per the same method as section 4.4.1. $L_S$ is the shadowing loss that may occur with environmental obstructions that may absorb, reflect or diffract signals. This term is universally modelled with a log normal distribution. $\zeta_{\text{LOS}}$ and $\zeta_{\text{NLOS}}$ are added as localisation variability and are modelled using the log normal distribution with 0 mean. All parameters in table 6 are obtained using empirical methods from [62] and shadowing parameters are dependent on frequency and elevation angle of each domain point, the latter of which was calculated using the same method as 4.4.1. For a robust design, we considered the scenario where each normally distributed parameter was set to a loss value two standard deviations above its mean. This approach ensured that there was only a 2.5% chance of experiencing a higher loss than what was accounted for in the design.

$$\begin{aligned} L_{\text{LOS}} &= \text{FSPL} + \zeta_{\text{LOS}} \\ L_{\text{NLOS}} &= \text{FSPL} + L_S + \zeta_{\text{NLOS}} \end{aligned} \qquad \text{where} \quad \begin{aligned} &L_S \sim \mathcal{N}(\mu_S, \sigma_S^2), \\ &\zeta_{\text{LOS}} \sim \mathcal{N}(\mu_{\text{LOS}}, \sigma_{\text{LOS}}^2), \\ &\zeta_{\text{NLOS}} \sim \mathcal{N}(\mu_{\text{NLOS}}, \sigma_{\text{NLOS}}^2) \end{aligned} \tag{12}$$

$$L_{\text{TOTAL}} = P(\text{LOS}) \times L_{\text{LOS}} + (1 - P(\text{LOS})) \times L_{\text{NLOS}} \tag{13}$$

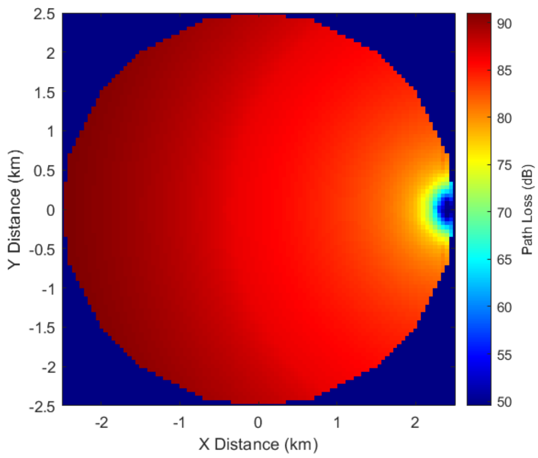


**Figure 9.** Trajectory 1 Losses

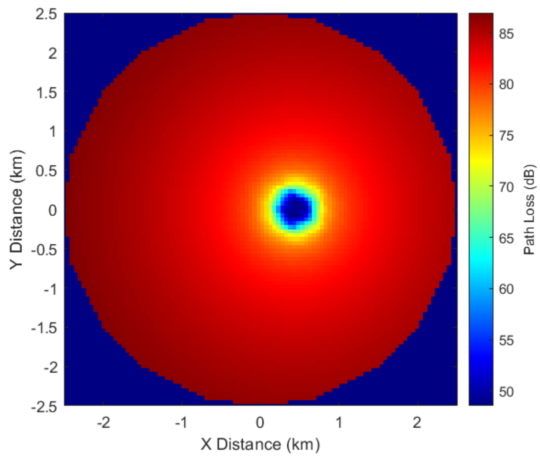


**Figure 10.** Trajectory 2 Losses

$L_{\text{TOTAL}}$ was calculated using equation (13) for each point in the domain for both trajectories. Similar to the communication system approach in Section 4.4.3, we considered design for the point in the domain with the worst value of $L_{\text{TOTAL}}$, which happens to be simply the furthest point from the drone's location (-2.5km,0km). The losses for the whole domain for both trajectories are shown in Figures 9 and 10. Atmospheric losses were neglected in this case due to being insignificant for transmission distances of this magnitude and being insignificant compared to shadowing losses.

**Table 6.** Statistical parameters used in Equation 12

| Parameter | Value |
|---|---|
| $\mu_S$ | $f(\beta, \text{frequency})$ |
| $\mu_{\text{LOS}}$ | 0 |
| $\mu_{\text{NLOS}}$ | 0 |
| $\sigma_S$ | $f(\beta, \text{frequency})$ |
| $\sigma_{\text{LOS}}$ | 5 |
| $\sigma_{\text{NLOS}}$ | 12 |

## 5.4 RSSI Calculation

RSSI is the parameter used to quantify the signal strength for ground users and is the standard parameter to design for in data relays and was calculated using equation (14) for the point (-2,5km,0km). Similar to section 4.4.3, the EIRP is varied between typical 4G data relay hardware values (10-40dBm) and the respective RSSI was calculated in figure 11. From literature a strong signal strength is equivalent to RSSI>-60dBm [63], thus this was designed for. This corresponded to EIRP of 27dBm and 31dBm for trajectory 1 and 2 respectively.

$$\text{RSSI} = \text{EIRP} - L_{\text{TOTAL}} \quad \text{where} \quad \text{EIRP} = \text{EIRP}_{\text{router}} + \text{G}_{\text{antenna}} \tag{14}$$

## 5.5 Data Rate and Capacity

Using equation (9) from section 4.4.3 with total available $B_W$=2GHz and the $\beta_{\max} = 1.33$bps/Hz for 4g LTE [54], the available bitrate was 2.67Gbps, accommodating approximately 26667 users. However, the bottleneck in our data relay system was the relay back to the GCS which occurs via the satellite terminal. From section 4.4.3 the available bitrate is 1.26Mbps for the satellite datalink. During the data relay stage where having surveillance is less important, the bitrate requirement of the low quality video feed is relaxed to 0.50Mbps and even lower quality video feed was used. Using the other parameters from table 3, this left 0.66Mbps for data relay. Assuming the 30kbps per person bitrate requirement from section 5.1, this accommodated approximately 22 people which under performed compared to the design requirements.

## 5.6 Hardware Selection

The most powerful routers suitable for UAVs typically have max $\text{EIRP}_{\text{router}}$ of 23dBm[1]. Furthermore, omni-directional and directional antennas typically have max $\text{G}_\text{t}$ of 3dBm and 8dBm respectively. Therefore, as a result of hardware constraints it was narrowly not possible to facilitate trajectory 2's requirement from section 5.4. The final router selected was the Teltonika RUTX09 ($\text{EIRP}_{\text{router}}$=23dBm) [64] and the directional antenna MIMO WMM86-7-27 ($\text{G}_\text{t}$=8dBm) [65] to be used in trajectory 1. These were selected due to having the most EIRP and $\text{G}_\text{t}$ while being significantly lighter than other similar performing hardware. The final combined mass for the system is 1.41kg and combined power 29W. As a directional antenna was used, in order to provide signal to

**Table 7.** Loss Contributions for Most Constraining Point

| Parameter | Value |
|---|---|
| $P(LOS)$ | 2.79e-13 |
| $FSPL$ | 48.53dB |
| $L_s$ | 18.91dB |
| $L_{\text{LOS}}$ | 59.38dB |
| $L_{\text{NLOS}}$ | 91.08dB |
| $L_{\text{TOTAL}}$ | 91.08dB |

[1] 1 dBm is equivalent to 0.001dBW

points directly below the UAV, the antenna itself must be angled towards the banking direction. Appendix E shows the arrangement based on the 1.8° trajectory bank angle and 100° vertical beam-width of the antenna. Design of the housing plate for the relay was performed by the Modular-Add-Ons subteam [66], using the previously mentioned hardware and alignment angle.

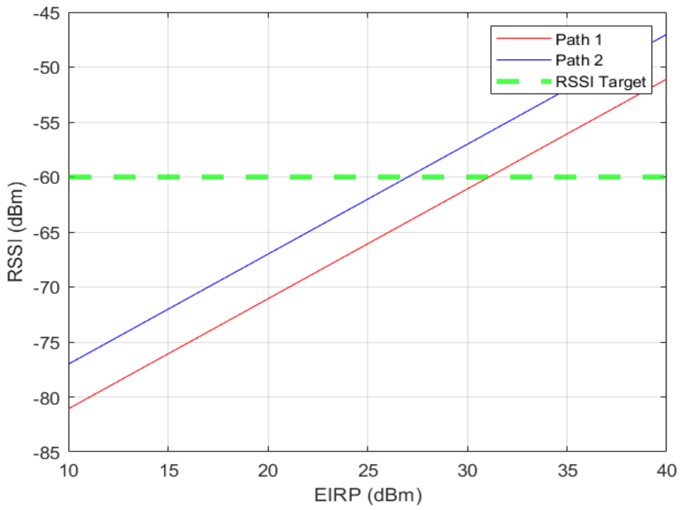


**Figure 11.** RSSI vs EIRP

# 6 Conclusions

In conclusion, this report has presented the metrics and trade-off studies used in order to select off-the-shelf Navigation sensors suitable for REACH driven by stringent criteria such as accuracy suitable for VTOL and environmental resistance. The decision was made to exclude hardware redundancies in favour of superior software-based fault management systems given recent advancements in software reliability. For future iterations, more in-depth market analysis will be conducted to select better performing sensors which was avoided in this iteration due to time constraints in this project.

In the Guidance system, the selection of the CNN YOLO for object detection, Octomap for terrain representation, RRT* for trajectory planning and Dubin's path for trajectory smoothing ensured a reliable obstacle detection and avoidance framework for autonomous flight given current technology that is suitable for REACH. Future development would include implementing the suggested workflow in section 3.5, using Unreal engine to validate the workflow and field tests to ensure reliability under diverse and challenging conditions. Perhaps with future advancements in more reliable machine learning algorithms, different parts of the frameworks can be updated with more accurate algorithms.

The Communication System was designed using SATCOM to meet the challenging 185km BLOS radius of communicating with the GCS. A subset of GEO IntelSat satellites were selected to facilitate this. The MILLI SAT terminal was selected following a novel design process modelling propagation and atmospheric losses using industry standard methods. By accounting for adverse rain conditions, the reliability of the UAV datalink was designed to be 99.9%. The calculated Link Margin and Bitrate was 8dB and 1.26Mbps respectively which successfully met the requirements for safety margin and data rates from section 4.1. For future iterations, investigation will be done into the use of LEO satellites that provide lower latency and higher data rates [67], allowing for use of lighter and less power intensive SATCOM terminals. This was avoided for this iteration due to the complicated nature of modelling changing orbits whilst ensuring global coverage.

The Data Relay system was far more challenging to design around the 185km mission radius. A similar novel design approach to the Communication system was taken, this time modelling urban losses to ensure successful coverage in the most challenging high rise urban environments and the most optimal path trajectories were considered. The final selected hardware was the Teltonika RUTX09 router and MIMO WMM86-7-27 directional antenna. As a result of the bottleneck of relaying data to the GCS using the SATCOM terminal the imposed requirement of allowing 50 people to send emergency data(100kbps) simultaneously was failed to be met, with only 22 people being facilitated. Although, this number would still significantly aid in data relaying operations, the target of 50 people could be met in the future with the previously mentioned use of LEO satellites, that would allow higher bandwidth allocations for data relay purposes. Perhaps in future iterations, dynamic bandwidth allocation can be used when relaying data via the satellite to the GCS to allow the existing bandwidth to facilitate more people [68].

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# Appendix

## A Navigation Sensor Market Analysis

Below are the typical sensor options and the respective states they measure. All symbols are consistent with meaning in typical flight dynamic analysis[7].

- GNSS ($x$, $y$, $z$)
- IMU ($u$, $v$, $w$, $\phi$, $\theta$, $\psi$, $p$, $q$, $r$)
- Magnetometer ($\psi$)
- Altitude Barometer ($z$)
- Airspeed Sensor ($u$)
- AoA wind vane ($u$, $w$)
- Ultrasonic sensor ($z$)
- Lidar ($x$, $y$, $z$)
- Optical Camera ($x$, $y$, $z$, $\phi$, $\theta$, $\psi$) [2]
- Infra-red Camera ($x$, $y$, $z$, $\phi$, $\theta$, $\psi$)
- Radar ($x$, $y$, $z$, $\phi$, $\theta$, $\psi$)

| Module Name | Accuracy (m) | Weight (g) | Size (mm) | Power (W) | Frequency (Hz) |
|---|---|---|---|---|---|
| Trimble BD9250 (Vib Resist) | 0.25-1.50 | **25** | 71.1 x 45.7 x 11 | **1.4** | **20** |
| AstreRx – m2 | **H: 1.2, V: 1.9** | **28** | **47.5 x 70 x 7.6** | **1.050** | **<20** |
| **Novatel OEM719** | **1.5** | **31** | **46x71x11** | **1.8** | **<100** |
| **Emlid Reach M2** | **0.1m** | **20** | **56.4x45.3x14.6** | **1** | **5** |
| **APX-15 UAV** | **1.5-3.0** | **60** | **67x60x15** | **3.5** | **<50** |
| **OEM7700** | **0.4-1.5** | **31** | **46x71x8** | **1.8** | **<100** |

**Figure 12.** GNSS Receiver Hardware Options

| **Receiver Name** | **Accuracy(m)** | **Weight(g)** | **Size(mm)** | **Power(W)** | **Frequency (Hz)** |
|---|---|---|---|---|---|
| **Novatel PwrPak7** | **0.4-1.5** | **500** | **147x125x55** | **3.25** | **<100 Hz** |
| **AsteRx-U3** | **0.4-1.9** | **1500** | **157x245x45** | **8** | **50Hz** |
| **Topcon hiperhr** | **0.003-0.01** | **1172** | **115x115x132** | **10** | **20** |

**Figure 13.** GNSS Antenna Hardware Options

| Name | Weight (g) | Dims (cm) | Gyro | | | Accel | | | Frequency (Hz) | Power (W) |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Bias Stab (°/hr) | Bias Repeat (°/hr) | Random Walk (°/√hr) | Bias Stab (mg) | Bias Repeat(mg) | Random Walk (m/s/√hr) | | |
| HG1120 | 54 | 4.7x4.39x1.41 | 10 | 260 | 0.3 | 0.03 | 5 | 0.04 | 100 | 0.4 |
| DMU41-02 | 170 | 5.05x5.05x5.1 | 0.1 | 17 | 0.02 | 0.015 | 2.3 | 0.05 | 200 | 2.5 |
| SBG Ellipse 2 Micro Series | 10 | 2.68x1.88x0.95 | 0.2 | 7 | 0.15 | 5 | 0.014 | 0.0035 | 100 | 0.4 |
| HG1125 | 24 | 3.1D 1,17H | 5 | 120 | 0.3 | 0.035 | 1.5 | 0.18 | 145 | 0.5 |

**Figure 14.** IMU Hardware Options

| Name | Mass(g) | Size(mm) | Accuracy (±m) | Resolution (±m) | Response Time (ms) | Power Consumption (μA) |
|---|---|---|---|---|---|---|
| DPS310 | <1 | 2.0x2.5x1.0 | 0.5 | 0.02 | 27.6 | 1.7 |
| BMP280 | <1 | 2.0x2.5x0.95 | 1 | 0.01 | 5.5 | 3.4 |
| BMP388 | <1 | 2.0x2.0x0.75 | 0.66 | 0.17 | 12.6 | 3.4 |
| MS5611-01BA03 | <1 | 5.0x3.0x1.0 | 0.5 | 0.1 | 1 | 1 |

**Figure 15.** Altitude Barometer Hardware Options

[2]Orientation is obtained through obstacle detection algorithms and used to obtain $\phi$, $\theta$, $\psi$

| Name | Size (cm) - Assume Cylinder | Weight (g) | Accuracy (±cm) | Angular Resolution (±°) | Range (m) | Scan Rate | Field of View (°) |
|---|---|---|---|---|---|---|---|
| Ouster Os1 | D: 8.5 H:7.35 | 447 | 3 | 0.01 | 100 | | 45 |
| Velodyne VLP-16 Puck HI-RES | D: 10.3 H: 7.2 | 830 | 3 | 0.1 | 100 | | 30 |
| benewake ce30-d | 8.3x5.7x5.4 | 334 | 15 | 0.19 | 30 | | 4 |
| Quanergy M8-1 | D:10.3 H:8.7 | 900 | 3 | 0.13 | 100 | | 20 |

**Figure 16.** Lidar Hardware Options

# B Obstacle Detection and Avoidance

| Name of the methods | | Pros | Cons | Common features |
|---|---|---|---|---|
| Cooperative Obstacle Sensing Methods | TCAS ADS-B FLARM | • High accuracy. | • Can only be used for UAVs equipped with the same system. | - |
| Visual Detection | Monocular Vision System | • Low cost. | • Affected by darkness or strong light. | • The accuracy is related to the performance of the image processing algorithm. |
| | Stereo Vision System | • Improved accuracy compared with monocular vision system. | • Affected by darkness or strong light.<br>• High cost. | |
| Non-visual Detection | Active Sensors | • Radar: Great detection range.<br>• Laser: High resolution.<br>• Ultrasonic Sensors: Low cost. | • Radar: Sensitive to electromagnetic interference.<br>• Laser: High cost.<br>• Ultrasonic Sensors: Sensitive to the meteorologic environment. | • The accuracy is related to the quality of sensors. |
| | Passive Sensors | • Low cost. | • Low accuracy. | |

**Figure 17.** Categories of detection hardware from [23]

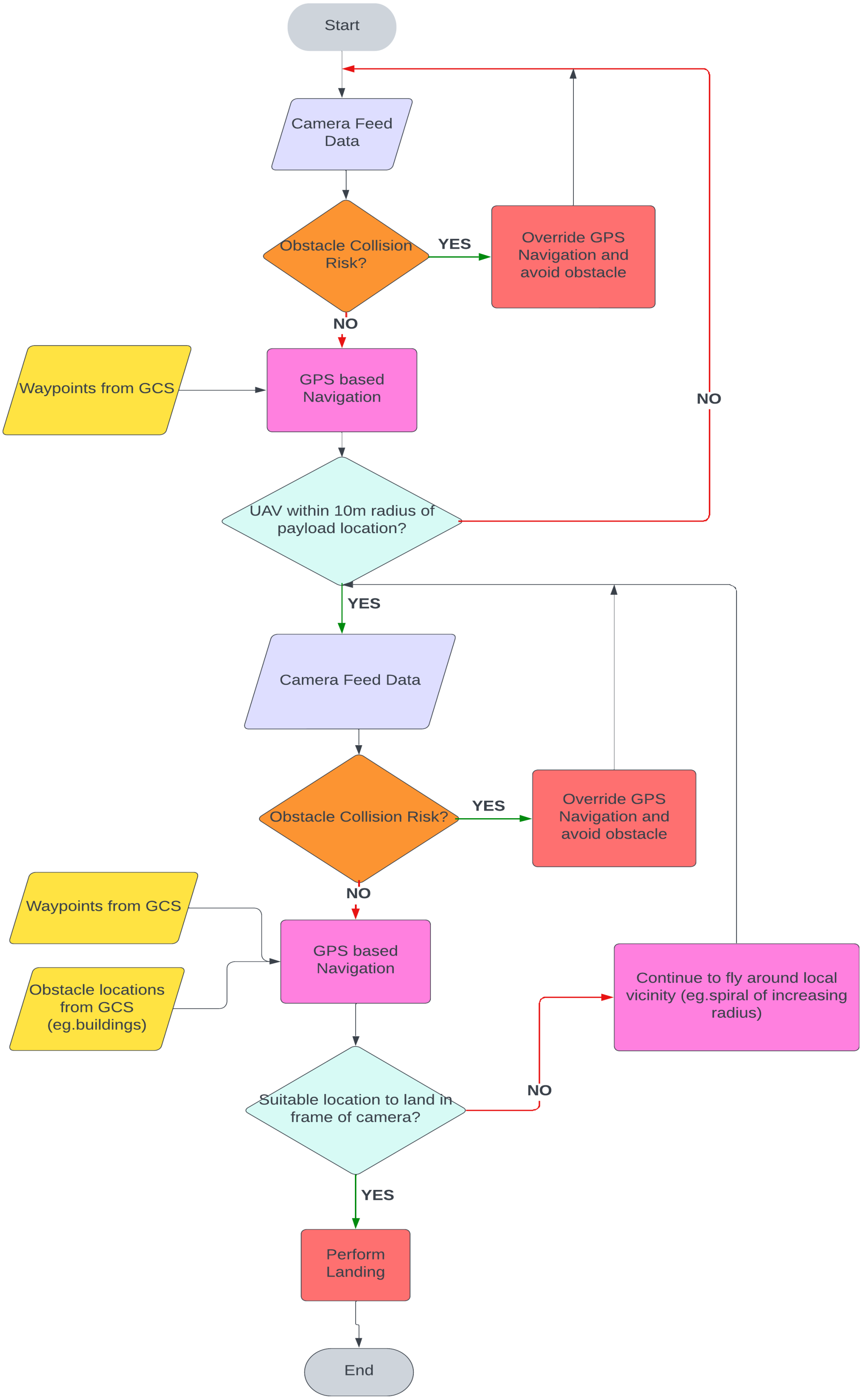


**Figure 18.** Example Payload Mission Guidance Algorithm

# C Satellite Fleet Data

**Table 8.** Chosen Satellite Fleet

| Satellite | Longitude °E | Coverage Area |
|---|---|---|
| Intelsat 17 | 66.0 | Africa, Europe, Middle East, Russia |
| Intelsat 18 | 180.0 | US, S.Pacific |
| Intelsat 20 | 68.5 | Africa, Central Asia, Europe, Middle East, Russia, S.Africa |
| Intelsat 33e | 60.0 | Eurasia |
| Intelsat 35e | 325.5 | Africa, Caribbean, Europe, Mediterranean |
| Intelsat 37e | 342.0 | Algeria |

# D Slant Range Calculation Geometry

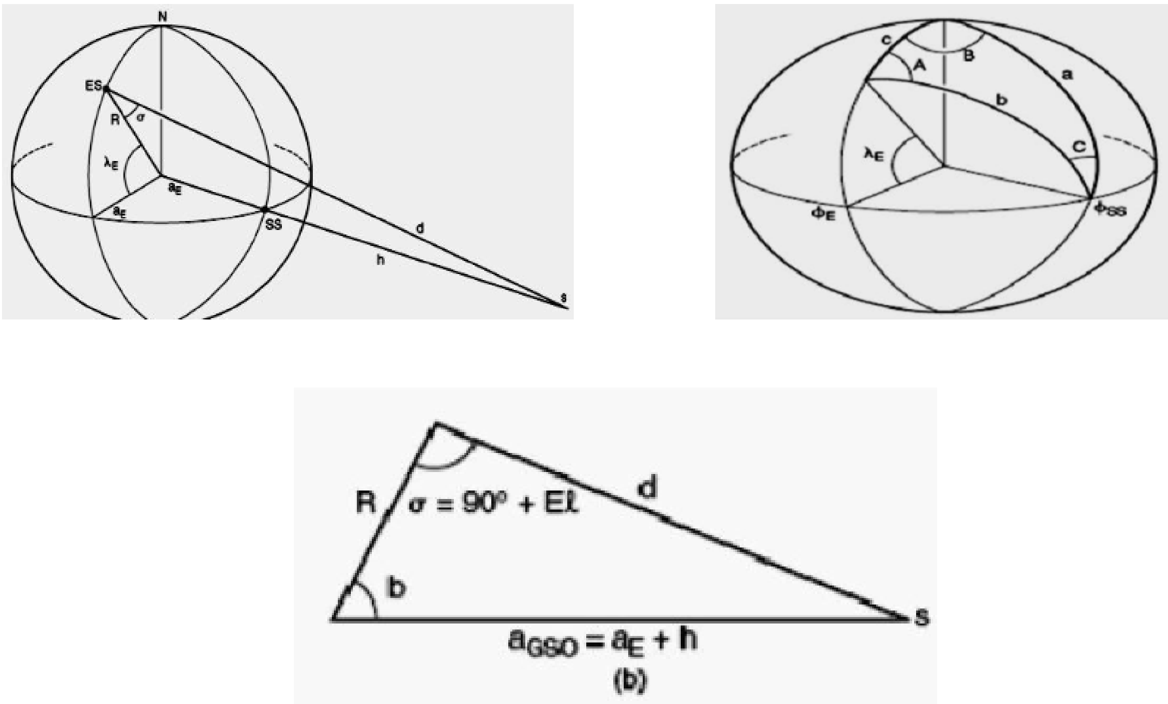


**Figure 19.** Geometry used to calculate slant range from [69]

# E Data Relay Antenna Configuration

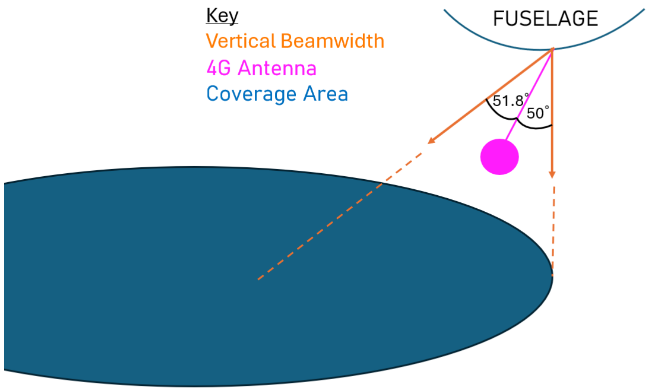


**Figure 20.** Data Relay Antenna angle